

Practical Example: Albatross Take-Off

Objective of the Example

Estimate the forces and moments (torques) acting on an albatross wing during take-off, using a simple mathematical model based on basic physics.

Problem Statement

When an albatross takes off, the air exerts a force on its wings. This force depends on:

- the airspeed,
- the size and shape of the wing,
- the angle at which the air hits the wing.

We want to estimate:

1. How much the air "pushes" the wing upward (lift force),
2. How much this force makes the wing want to bend (bending moment),
3. How much this force tries to twist the wing (torsional moment).

Known Data

- Wing semi-span: $b/2 = 1$ m
- Root chord (width at the center): $c_r = 0.2$ m
- Chord distribution: $c(z) = c_r \sqrt{1 - 4(z/b)^2}$
- Airspeed: $U_\infty = 15$ m/s
- Air density: $\rho = 1.225$ kg/m³
- Angle of attack: $\alpha = 10^\circ = 0.1745$ rad
- Lift coefficient: $C_L = 2\pi\alpha \approx 1.096$
- Drag coefficient: $C_D = 0.04$
- Elastic axis – aerodynamic center offset: $x_{EA} - x_{AC} = -0.02$ m

Step 1: Forces on the Wing

The aerodynamic pressure on the wing per unit area is:

$$F = \frac{1}{2}\rho U^2$$

This pressure is multiplied by the wing area and the lift coefficient C_L to get the upward lift force:

$$F_L(z) = \frac{1}{2}\rho U^2 c(z) C_L$$

We also break this force into two components:

- C_N : normal to the wing surface.
- C_X : along the chord of the wing.

Calculations:

$$C_N = C_L \cos \alpha - C_D \sin \alpha \approx 1.071$$

$$C_X = C_L \sin \alpha + C_D \cos \alpha \approx 0.229$$

Step 2: Bending Moment (Flapwise Moment)

The further from the root, the larger the moment arm. The bending moment is:

$$M_F = \int_0^{b/2} \left(\frac{1}{2}\rho U^2 c(z) C_N \right) z dz$$

With an elliptical chord distribution, this becomes:

$$M_F = \frac{c_r b^2}{24} \rho U^2 C_N$$

Substituting:

$$M_F = \frac{0.2 \cdot (2)^2}{24} \cdot 1.225 \cdot 15^2 \cdot 1.071 \approx \boxed{9.84 \text{ Nm}}$$

Step 3: Edgewise Moment

This is the aerodynamic moment along the flight direction:

$$M_E = \frac{c_r b^2}{24} \rho U^2 C_X$$

$$M_E \approx \frac{0.2 \cdot 4}{24} \cdot 1.225 \cdot 225 \cdot 0.229 \approx \boxed{2.10 \text{ Nm}}$$

Step 4: Torsional Moment

Due to the offset between the aerodynamic force center and the structural torsion axis:

$$M_T = \frac{\pi c_r b}{16} \rho U^2 C_R (x_{EA} - x_{AC})$$

Where:

$$C_R = \sqrt{C_L^2 + C_D^2} \approx 1.097$$

Substituting:

$$M_T = \frac{\pi \cdot 0.2 \cdot 2}{16} \cdot 1.225 \cdot 225 \cdot 1.097 \cdot (-0.02) \approx \boxed{-3.55 \text{ Nm}}$$

The negative sign indicates the twist is opposite to the reference direction.

Conclusion

During take-off, the albatross wing experiences:

- A strong bending moment: 9.84 Nm
- A smaller edgewise moment: 2.10 Nm
- A significant torsional moment: -3.55 Nm

This explains why wings must be very stiff, especially in torsion, to avoid dangerous vibrations or excessive deformation during flight.